Viswa Sushanth Karuturi

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EDUCATION

University of Arizona, College of Science — **Tucson, Arizona**
Bachelor of Science in Computer Science — December 2026
Bachelor of Science in Statistics & Data Science
Minor: Information Science, Technology & Arts

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C/C++, Bash/Shell scripting
Data Engineering: Data Ingestion, Data Cleaning, Data Transformation, ETL Workflows, Structured Data, Data Quality
Databases & Cloud: AWS S3, Amazon DynamoDB, PostgreSQL, Data Storage & Retrieval
Tools & Analytics: Pandas, NumPy, Git/GitHub, Linux, Docker, Streamlit, Tableau

TECHNICAL EXPERIENCE

VIP - Getting to Know U — **Tucson, Arizona**
Undergraduate Research Assistant — January 2026 – May 2026
- Developed AWS S3 and DynamoDB storage and retrieval workflows for structured academic data supporting a university degree-planning platform.
- Built cloud-based data ingestion and transformation workflows to prepare academic datasets for analytics, dashboards, and predictive modeling.

STAR-X Club – NASA Lunabotics Competition — **Tucson, Arizona**
Software & Autonomy Lead — November 2025 – Present
- Lead ROS-based software development for a NASA Lunabotics rover, integrating navigation, control, and telemetry components.
- Coordinate software integration with mechanical and electrical teams to support reliable testing and rover operation.

University of Arizona, Department of Computer Science — **Tucson, Arizona**
Undergraduate Teaching Assistant — January 2025 – May 2025
- Supported 100+ students in discrete mathematics, strengthening skills in proofs, graphs, trees, and analytical problem-solving.
- Designed and led structured review sessions explaining technical concepts and mathematical problem-solving techniques.

PROJECTS

Tucson Crime Insights | *Python, scikit-learn, Streamlit, GeoJSON* | GitHub | Dashboard
- Processed 243,127 Tucson crime records (2018–2025), cleaning missing values, parsing dates, and engineering structured features for downstream analysis.
- Integrated cleaned records with GeoJSON geographic boundary data and built a Streamlit dashboard to explore crime patterns by division and ward.

Dynamic Pricing using Reinforcement Learning | *Python, Reinforcement Learning* | GitHub
- Built a dynamic pricing simulation with finite inventory, stochastic demand, and a defined sales horizon for repeatable computational experiments.
- Structured simulated inventory, customer demand, and pricing state variables to evaluate sequential decisions and revenue-related outcomes.

Supreme Court Case Outcome Predictor | *Python, Machine Learning, Neural Networks* | GitHub
- Prepared structured case data for classification, applying preprocessing and feature representation techniques to support model development.
- Explored feature embeddings and feedforward neural networks to compare representations of case attributes for outcome prediction.

Financial Collapse Indicators | *R, Statistical Modeling, Regression* | GitHub
- Prepared approximately 6,000 cross-country observations (1990–2023) for reproducible statistical modeling and economic indicator analysis.
- Applied data transformations, influential-observation checks, and multicollinearity diagnostics to evaluate dataset quality and model robustness.

PROFESSIONAL & LEADERSHIP EXPERIENCE

University of Arizona, College of Information Science — **Tucson, Arizona**
Front Office & Business Operations Student Worker — August 2024 – Present
- Support business operations through organized financial documentation, purchasing-card reconciliation, and reimbursement processing.

Tucson Meet Yourself — **Tucson, Arizona**
Volunteer Leadership — October 2023 – October 2026
- Progressed from volunteer to Area Captain and Assistant Training Manager, supporting volunteer coordination and festival teams.
- Developing a volunteer training manual covering festival sections, responsibilities, and operational procedures.

CERTIFICATIONS & ACTIVITIES

Certifications & Training: Python for Everybody (University of Michigan), Python for Data Science, AI & Development (IBM), Agile with Atlassian Jira, Data Visualization with Tableau
Activities: STAR-X Club; Community Volunteering